On epidemiological models and their tropical approximation

Mohamed Amine Rabhi and Reda Belhaj-Soullami , under the supervision of Stéphane Gaubert

Abstract

We study in this paper the problem of approximating epidemiological models, as formulated in [4]. We focus on the tropicalization of models such as SIR and SEIR in order to predict the impact of sanitary models on the evolution of the Covid-19 pandemic. This has led us to study the time of peak of these models, which we found a satisfactory approximation for the SIR model thanks to [1]. We then brought to light some results that might give the same kind of approximation for the SEIR model and multi-compartment models.

1 Introduction

The Covid 19 outbreak has reinvigorated the study of simple and multi-compartment epidemiological systems. These systems, mainly derived from the work of Kermack and McKendrick [5], are still poorly understood because they are described by systems of non-linear equations.
Indeed, the SIR and SEIR models, although simple in conceptualization, lead to the use of complex tools when we want to characterize their properties or approximate their dynamics. For instance, among these tools, tropical geometry has led to promising results in term of approximating the early stage of an epidemic. [4] shows that this early stage can be approximated by an exponential function, that can be obtained after tropicalyzing the SIR system.

In this paper, we started from the results of [4] in order to study more deeply the link between tropical geometry and the approximation of such systems. We show in 2 that the SIR model has two main exponential behaviors which can be brought to light after calculating the amoeba of the system. We generalize this property to the bi-compartment model and study the main features of these amoebae. In term of approximation, the work of [4] shows that the time of peak is the moment of switching between the two exponential behaviors. Therefore, we naturally focused on approximating this time in order to represent SIR epidemics by only two functions. Thanks to the work of Bacaër in [1], we developed a fairly satisfactory approximation which was later confirmed by Bacaër in [2]. We try to generalize theses approximations to SEIR or bi-compartment models, but only managed to obtain intermediary results.

Along all this paper, we will make reference to SIR, SEIR and bi-compartment models. For the sake of being consistent in our notations, we list below the three differential systems as well as their significance.

1.1 SIR model

$$\begin{cases} \frac{dS}{dt} = -k_{I,S}\frac{S}{N}I \\ \frac{dI}{dt} = k_{I,S}\frac{S}{N}I - k_{I,R}I \\ \frac{dR}{dt} = k_{I,R}I \end{cases} \tag{1}$$

In the SIR model, N represents the size of the population (which is supposed to remain constant), S the number of susceptible individuals, I the number of infectious individuals, and R for the number of removed (deceased or immune) individuals. Likewise, $k_{I,S}$ and $k_{I,R}$ represent the share of the susceptible population that gets the decease at each moment and the share of the infectious population

that is removed at each moment respectively. We will always take $S(0) = N - 1$, $I(0) = 1$ and $R(0) = 0$. Giving the system of equation above, we therefore always have $S + I + R = N$. Here's a typical graph of the evolution of each compartment along the time:

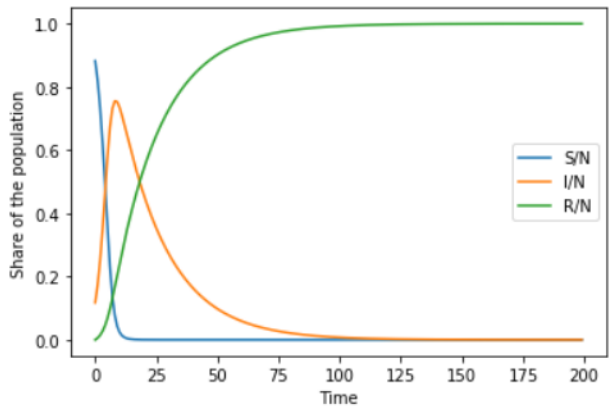

Figure 1: SIR evolution graph

1.2 SEIR model

$$\begin{cases} \frac{dS}{dt} = -k_{I,S}\frac{S}{N}I \\ \frac{dE}{dt} = k_{I,S}\frac{S}{N}I - k_{E,I}E \\ \frac{dI}{dt} = k_{E,I}E - k_{I,R}I \\ \frac{dR}{dt} = k_{I,R}I \end{cases} \tag{2}$$

In the SIR model, N represents the size of the population (which is supposed to remain constant), S the number of susceptible individuals,E the number of exposed I the number of infectious individuals, and R for the number of removed (deceased or immune) individuals. Likewise, $k_{I,S}$, $k_{E,I}$ and $k_{I,R}$ represent the share of the susceptible population that gets exposed to the decease at each moment, the share of the exposed population that gets the decease at each moment, and the share of the infectious population that is removed at each moment respectively. We will always take $S(0) = N - 1$,$E(0) = 0$ $I(0) = 1$, and $R(0) = 0$. Giving the system of equation above, we therefore always have $S + E + I + R = N$. Here's a typical graph of the evolution of each compartment along the time:

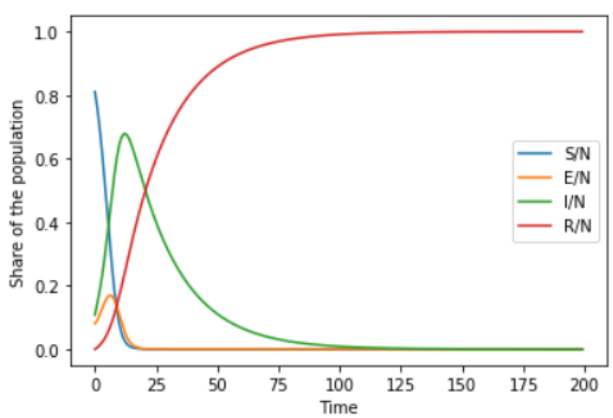

Figure 2: SEIR evolution graph

1.3 Bi-compartment SIR model (no transport assumption)

$$\begin{cases} \frac{dS_1}{dt} = -\beta_{1,1}\frac{S1}{N1}I_1 - \beta_{1,2}\frac{S1}{N1}I_2 \\ \frac{dS_2}{dt} = -\beta_{2,1}\frac{S2}{N2}I_1 - \beta_{2,2}\frac{S2}{N2}I_2 \\ \frac{dI_1}{dt} = \beta_{1,1}\frac{S1}{N1}I_1 + \beta_{1,2}\frac{S1}{N1}I_2 - \gamma_1 I_1 \\ \frac{dI_2}{dt} = \beta_{2,1}\frac{S2}{N2}I_1 + \beta_{2,2}\frac{S2}{N2}I_2 - \gamma_2 I_2 \\ \frac{dR1}{dt} = \gamma_1 I1 \\ \frac{dR2}{dt} = \gamma_2 I2 \end{cases} \tag{3}$$

In this model, N_1 represents the size of the first population (which is supposed to remain constant) and N_2 the size of the second one.
S_i the number of susceptible individuals, I_i the number of infectious individuals, and R_i for the number of removed (deceased or immune) individuals of the i-th population ($i \in \{1, 2\}$). Likewise, $(\beta_{i,j})_{i,j\in\{1,2\}}$, $k_{E,I}$ and $(\gamma_i)_{i\in\{1,2\}}$ represent the share of the i-th susceptible population that gets the decease at each moment due to contact with the j-th population and the share of the i-th infectious population that is removed at each moment respectively. We will always take $S_i(0) = N - 1$,$I_i(0) = 1$, and $R_i(0) = 0$. Giving the system of equation above, we therefore always have $S_i + I_i + R_i = N_i$ for each $i \in \{1, 2\}$.
Usually, the bi-compartment model contains transport terms, in order to describe the epidemiological dynamics between two geographically distinct populations, e.g. between two towns. Here, the model we've chosen is rather describing what would happen if two close populations were infected, e.g. old and young people in the same geographic area.
Here's a typical graph of the evolution of each compartment along the time:

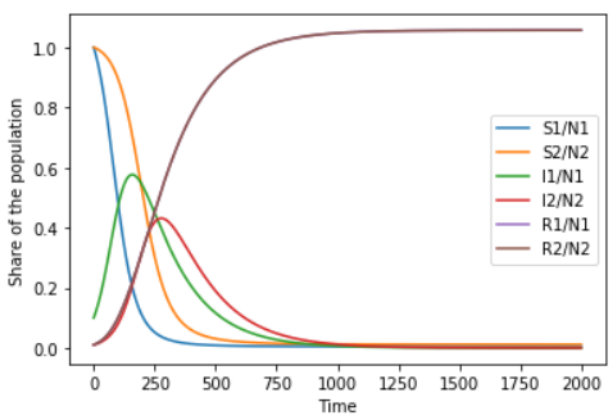

Figure 3: Bi-compartment evolution graph

2 Tropical approximation for epidemiological differential systems

In this section, we present a technique for approximating the solutions of a differential system and apply it to the aforementioned epidemiological models. This technique has been used in [6] in the case of equations for biochemical reactions.

2.1 General principle

We are interested in approximating the differential system

$$\frac{\mathrm{d}x_i}{\mathrm{d}t} = P_i(\mathbf{x})$$

where $1 \leq i \leq n$, $\mathbf{x} = (x_1, \ldots, x_n)$ and P_i is a multivariate polynomial : $P_i(\mathbf{x}) = \sum_{\alpha\in A_i} a_{i,\alpha}\mathbf{x}^\alpha$.
Following [6] we call "tropicalization of the system" the following system :

$$\frac{\mathrm{d}x_i}{\mathrm{d}t} = s_i \exp\left(\max_{\alpha \in A_i}(\log|a_{i,\alpha}| + \langle \mathbf{u}|\alpha\rangle)\right)$$

where $\mathbf{u} = \log(x)$ is the point-wise logarithm of vector $\mathbf{x}$, and s_i is the sign of $a_{i,\alpha}$ where α realizes the maximum.

In this piecewise approximation, only one dominant term is kept on the RHS of the differential system. It is hence necessary to compute the set of points $\mathbf{u}$ that delimit the domains where each term is dominant. These points are exactly the points where the maximum is attained at least twice. This set is called the ameoba of the system.

2.2 Tropical approximation of the SIR model

In the case of the SIR model, represented by Equation 1, the ameoba is easy to compute. We denote by $\bar{S}, \bar{I}, \bar{R}$ the logarithms of S, I, R. Only the second equation has more than one term. The associated tropical polynomial is

$$\max(\log(k_{I,S}/N) + \bar{S} + \bar{I}, \log(K_{I,R}) + \bar{I})$$

The equation for the ameoba is then $\bar{S} = \log(NK_{I,R}/K_{I,S})$: the amoeba is just a vertical line in a $(\bar{S}, \bar{I})$ plane.

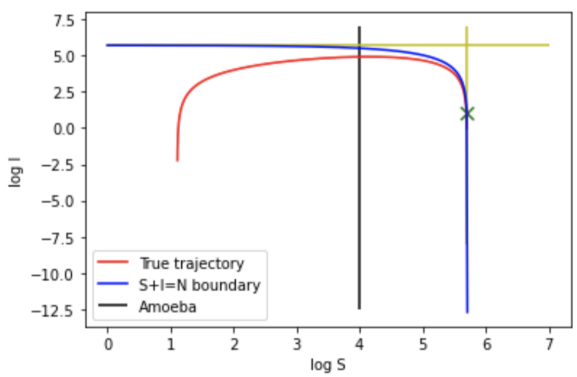

Figure 4: SIR amoeba

In Figure 4 the amoeba is represented as a black line. The true trajectory in the phase space (i.e. the set of points $(\log I(t), \log S(t))_{t\geq 0}$ is represented in red. The epidemic starts on the left (green cross). In a first phase, the number of infected people grows rapidly and the number of susceptible decreases slowly. In a second phase, after reaching a peak, the number of infected people slowly decreases as the number of susceptible people decreases rapidly. Finally the number of infected people decreases to zero, as the number of susceptible people converges to S_∞, the number of people who where not infected by the epidemic. The two phases for I - before and after the peak - will be approximated with the amoeba and the time of peak in the next sections.

2.3 Tropical approximation of the bi-compartment model

In the case of the bi-compartment model, represented by Equation 3, the amoeba is more difficult to compute. Indeed, giving the fact that R_1 ans R_2 do not impact the global behavior of the system, the amoeba lives in a 4-dimensional space: $(\bar{S}_1, \bar{S}_2, \bar{I}_1, \bar{I}_2)$. If we focus on the fist equation, the tropical polynomial is:

$$\max(\log(\beta_{1,1}) - \log(N_1) + \log(\bar{S}_1) + \log(\bar{I}_1), \log(\beta_{1,2}) - \log(N_1) + \log(\bar{S}_1) + \log(\bar{I}_2))$$

Therefore, the equation of the part of this amoeba is: $\bar{I}_1 - \bar{I}_2 = \log(\frac{\beta_{1,1}}{\beta_{1,2}})$. We can keep calculating the different equations characterizing the amoeba. We obtain a set of 3-dimensional sub-spaces with a non-zero intersection. We manage to plot the intersection of the amoeba with the space $(O, \bar{S}_2, \bar{I}_1, \bar{I}_2)$. Here's some screenshot of the set:

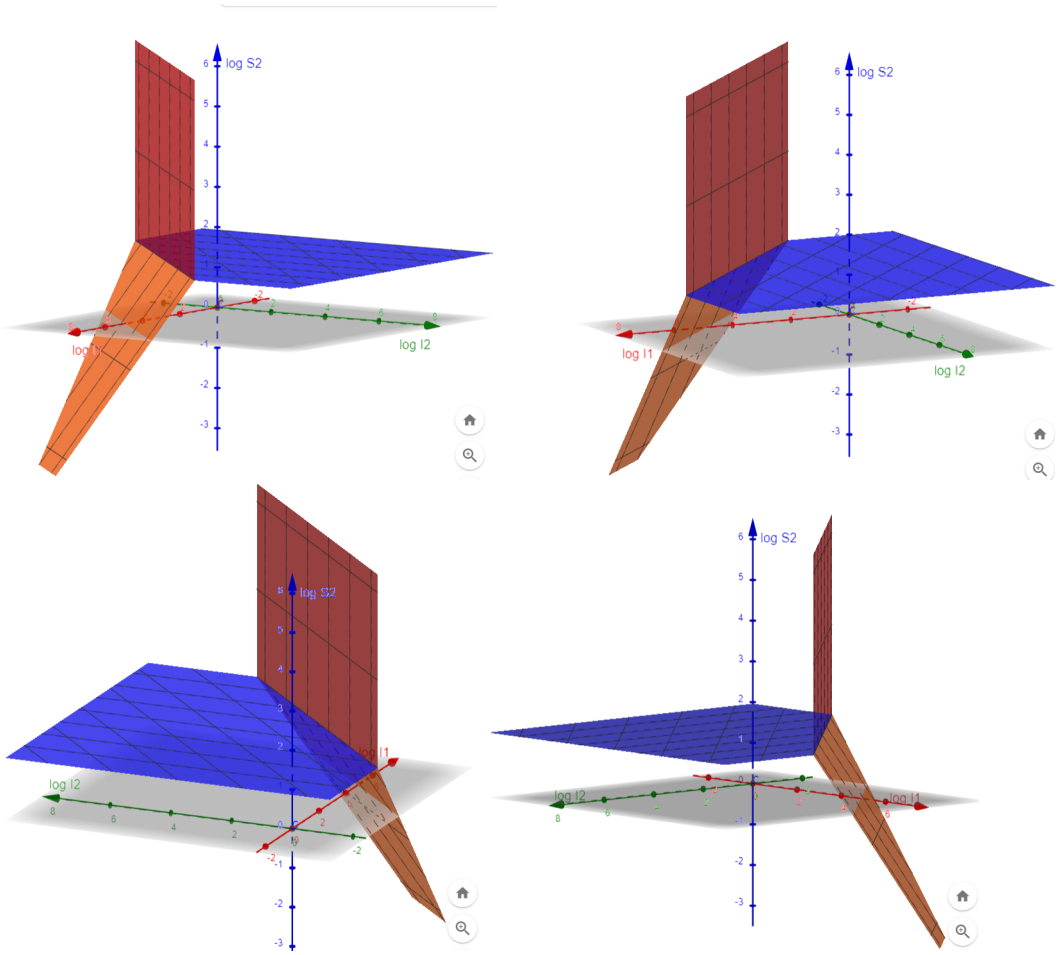

Figure 5: Intersection with the amoeba and the $(O, \bar{S}_2, \bar{I}_1, \bar{I}_2)$ space

In the set of figures above 5, we can imagine what will be the trajectory of the system given the plots 3. Indeed, we know that we start at the point $S_2 = N_2 - 1, I_1 = I_2 = 1$, therefore we start high in the blue axis then we should cross perpendicularly the blue plane, then the orange one and eventually the red one. At the end, we should expect to be at $-\infty$ in all the directions as all the population should be gathered in the $R_1 \cup R_2$ compartment.

In each of the cells delimited by the amoebae, the trajectory seems to be a line in the log-space. This was confirmed by the work of [4]. Therefore, in each cell, one can fairly approximate the trajectory by a line, which is surprisingly precise giving the figure 6 below. Yet, the problem of the moment of crossing each cell - and therefore the moment of passing from one slope to another - remains unknown. Thus, we naturally focused on approximating this moment, which is exactly the time of peak for the SIR model.

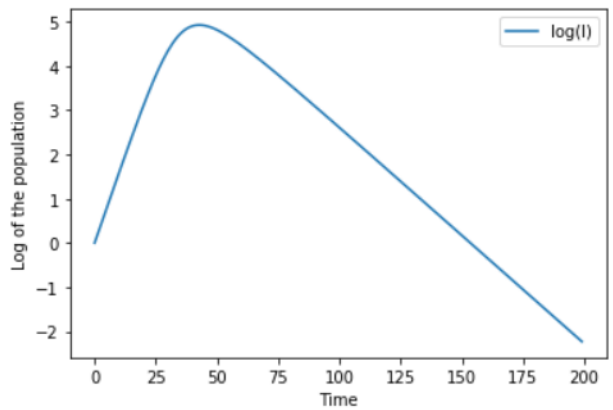

Figure 6: $\bar{I}$ with its two main branches

3 Time of peak approximations

In this section, we are interested in computing approximations for the time of peak for various models. Indeed, in order to construct a piecewise linear approximation of the system, as the slopes are given by the tropical polynomial, we need to approximate the time where a change of regime occurs, or the value of the observable at this time.

3.1 Approximating the time of peak in the SIR case

For approximating the time of peak in this case, we build upon Bacaer's work [1]. We will take the following initial conditions :

$$S(0) = N - 1, I(0) = 1, R(0) = 0$$

3.1.1 Refinement of Bacaër's bound

By combining the first and the last equation, it is easy to show that

$$\frac{1}{S}\frac{\mathrm{d}S}{\mathrm{d}t} = -\frac{K_{I,S}}{K_{I,R}N}\frac{\mathrm{d}R}{\mathrm{d}t}$$

and integrating with the initial conditions yields

$$\log\left(\frac{S(t)}{N-1}\right) = -\frac{K_{I,S}}{K_{I,R}N}R(t)$$

Plugging back the expression of $R(t)$ in the first equation gives the following independent equation for S:

$$\frac{\mathrm{d}S(t)}{\mathrm{d}t} = -\frac{K_{I,S}}{N}S(t)\left(N - S(t) + \frac{NK_{I,R}}{K_{I,S}}\log\left(\frac{S(t)}{N-1}\right)\right)$$

Denote by T the time of peak of the epidemic : $I(T) = \sup_{t\geq 0} I(t)$.

As S is nonincreasing, one can use a change of variables to compute the time of peak, provided the value of $S(T)$ is known. The second equation gives $S(T) = N\frac{K_{I,R}}{K_{I,S}}$. The change of variables gives :

$$T = \int_0^T \mathrm{d}t = \int_{N-1}^{S(T)} \frac{\mathrm{d}S}{-\frac{K_{I,S}}{N}S(t)\left(N - S(t) + \frac{NK_{I,R}}{K_{I,S}}\log\left(\frac{S(t)}{N-1}\right)\right)}$$

Let $s = S/N$. Then,

$$T = \frac{1}{K_{I,S}} \int_{K_{I,R}/K_{I,S}}^{1-1/N} \frac{\mathrm{d}s}{s(1 - s + \frac{K_{I,R}}{K_{I,S}} \log(\frac{s}{1-\frac{1}{N}}))}$$

Next, we compute an approximation of $T(N)$ as $N \to \infty$. First, we compute an upper bound for $T(N)$ using the fact that $\log(\frac{s}{1-1/N}) \geq \log(s)$.

Let $\alpha = \frac{K_{I,R}}{K_{I,S}}$. Then we have the upper bound $T(N) \leq \frac{1}{K_{I,S}} c(N)$ with

$$c(N) = \int_{\alpha}^{1-1/N} \frac{\mathrm{d}s}{s(1 - s + \alpha \log(s)}.$$

Let $g_N(s) = \frac{\mathrm{d}s}{s(1-s+\alpha \log(s)}$. A Taylor expansion at $s = 1$ shows that $g_N(s) = \frac{1}{(1-\alpha)(1-s)} + O(1)$. Hence the integral is divergent, and we can integrate the equivalent.

$$\int_{\alpha}^{1-\frac{1}{N}} \frac{\mathrm{d}s}{(1-\alpha)(1-s)} = \frac{1}{1-\alpha}(\log(N) + \log(1-\alpha))$$

Hence $c(N) = \frac{\log N}{1-\alpha} + o(\log(N))$ and $T(N) \leq \frac{\log(N)}{K_{I,S}-K_{I,R}} + o(N)$.

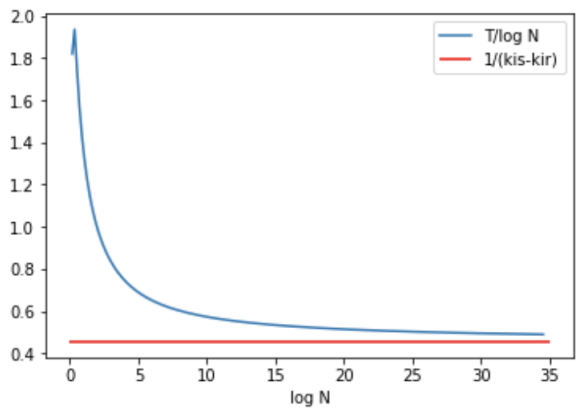

Figure 7: Convergence of the $O(\log(N))$ estimation of the time of peak

In Bacaër's paper [1], the lower bound is proven, giving $T(N) = \frac{\log(N)}{K_{I,S}-K_{I,R}} + o(\log(N))$. A visualization of the convergence of the estimate is shown in Figure 7. However, for our purposes, having an approximation up to $\log(N)$ precision is not enough. We continue this type of calculations to gain more precision. Write

$$d(N) = c(N) - \int_{\alpha}^{1-1/N} \frac{\mathrm{d}s}{(1-\alpha)(1-s)} = \int_{\alpha}^{1-1/N} g(s)\mathrm{d}s$$

With $g(s) = \frac{1}{s(1-s+\alpha \log s)} - \frac{1}{(1-\alpha)(1-s)}$.

A new Taylor expansion at $s = 1$ gives

$$g(s) = -\frac{K_{I,S} - 2}{2(K_{I,S} - 1)^2} + o(1)$$

This function is continuous at $s = 1$. Let $\varphi(\alpha) = \int_{\alpha}^{1} g(s)\mathrm{d}s$. We have $\varphi(\alpha) = d(N) + \int_{1-1/N}^{1} g(s)\mathrm{d}s$.

Denote by $e(N)$ this last integral. As g is continuous the integral is convergent and the remainders are equivalent, which shows that $\varphi(\alpha) = d(N) + O(1/N)$. Finally,

$$c(N) = \varphi(\alpha) + \frac{1}{1-\alpha}(\log(N) + \log(1-\alpha)) + O(1/N)$$

Then :

$$T(N) \leq \frac{\log N}{K_{I,S} - K_{I,R}} + \frac{\log(1-\alpha)}{K_{I,S} - K_{I,R}} + \frac{\varphi(\alpha)}{K_{I,S}} + O(1/N)$$

We managed to improve Bacaër's bound from $O(\log(N))$ to $O(1/N)$, giving a more precise approximation of the time of peak that will be useful for the piecewise linear approximation of the trajectory.

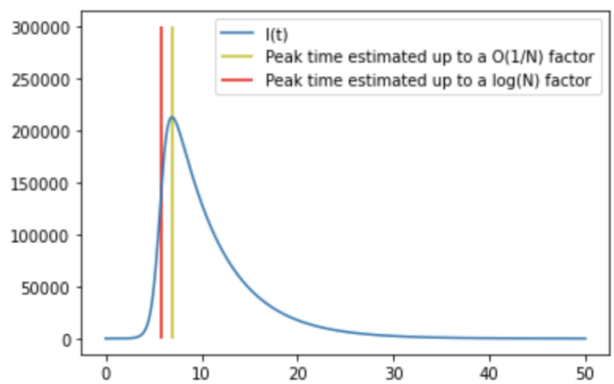

Figure 8: A plot showing the improvement on the estimation of the peak time

3.1.2 Properties of $f(\alpha)$

Now let's give a few qualitative properties of $f(\alpha)$. Recall that the $O(1)$ factor in the time of peak approximation is $\frac{1}{K_{I,S}}(\frac{\log(1-\alpha)}{1-\alpha} + \varphi(\alpha))$. Recall also that for a non trivial epidemic one has $K_{I,S} > K_{I,R}$ which is equivalent to $\alpha < 1$.

We now plot the $O(1)$ term w.r.t the parameter α.

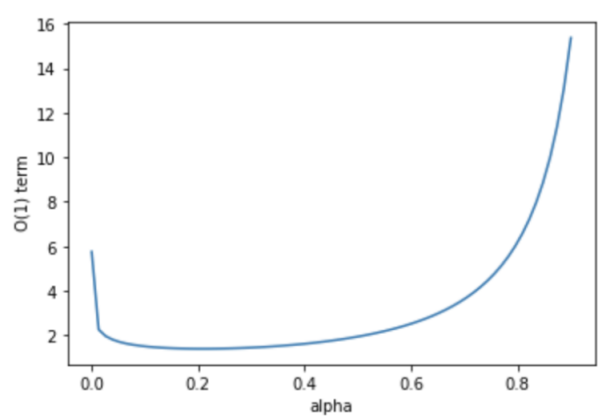

Figure 9: A plot of the $O(1)$ term w.r.t. α

3.1.3 Tropical approximation of the SIR model

Now we seek a piecewise linear approximation of $\log I$, using both the amoeba and the peak time approximation. We will use the notation $\tilde{S}, \tilde{I}, \tilde{R}$ for the approximations of S, I, R.

In the first phase of the epidemic (from $t = 0$ to $t = T$ the time of peak), we use the approximation $\tilde{S}/N \simeq 1$, which is valid if K_{IR}/K_{IS} is not too small. During this phase, the equation for $\tilde{I}$ becomes

$$\frac{\mathrm{d}\tilde{I}}{\mathrm{d}t} = (K_{I,S} - K_{I,R})\tilde{I}$$

Hence

$$\forall t \in [0, T], \tilde{I}(t) = e^{(K_{I,S} - K_{I,R})t}$$

After the peak, the tropical approximation reads

$$\frac{\mathrm{d}I}{\mathrm{d}t} = -K_{I,R} I(t)$$

hence

$$\forall t \geq T, \tilde{I}(t) = c + e^{-K_{I,R} t}$$

Then, using the approximation of the T, it is possible to deduce a piecewise linear approximation for $\log I(t)$ that is continuous. However, we found that using a non continuous approximation (as well as the expression of $I(T)$) fits the true trajectory better.

As $S(T) = N \frac{K_{I,R}}{K_{I,S}}$ is already known, one can use the equation

$$\log\left(\frac{S(t)}{N-1}\right) = -\frac{K_{I,S}}{K_{I,R} N} R(t)$$

to deduce that

$$I(T) = N\left(1 - \alpha + \alpha\log(\alpha) + \alpha\log\left(1 + \frac{1}{N-1}\right)\right) = N(1 - \alpha + \alpha\log(\alpha)) + o(1)$$

We use this expression to deduce the value of c in the expression of $\tilde{I}(t)$ for $t \geq T$.

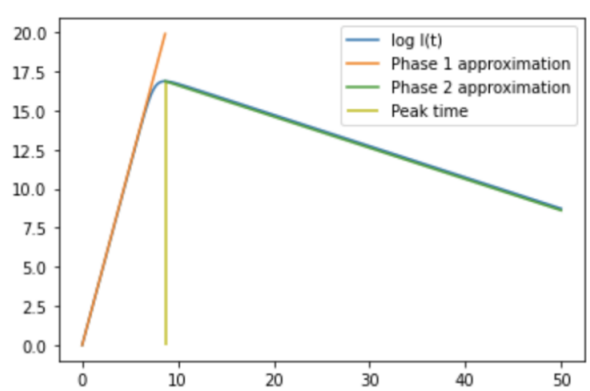

Figure 10: Tropical approximation of the SIR model

3.2 Approximating the time of peak in the bi-compartment case

In this section, let's focus on doing the same kind of development as we did for the SIR model for the bi-compartment one. We want to find an approximation of the first time of peak of the model. We start by finding independent equations verified by S_1 and S_2. By using the same method than before, we have:

$$\begin{pmatrix} \log(\frac{S_1(t)}{S_1(0)}) \\ \log(\frac{S_1(t)}{S_1(0)}) \end{pmatrix} = -\begin{pmatrix} \beta_{1,1}/N_1 & \beta_{1,2}/N_2 \\ \beta_{2,1}/N_1 & \beta_{2,2}/N_2 \end{pmatrix} \begin{pmatrix} \gamma_1 & 0 \\ 0 & \gamma_2 \end{pmatrix}^{-1} \begin{pmatrix} R_1(t) \\ R_2(t) \end{pmatrix}$$

Let's call B and Γ the two matrix above. Then, by injecting the relation $I_i = N_i - S_i - R_i$ into the second equation, we have:

$$\frac{d}{dt}\begin{pmatrix} \bar{S}_1 \\ \bar{S}_2 \end{pmatrix} = -B(N - S - R)$$

with N, S and R the vectors of each population. We therefore conclude that:

$$\frac{d}{dt}\bar{S} = -B(N - S) - \Gamma \begin{pmatrix} \log(\frac{S_1(t)}{S_1(0)}) \\ \log(\frac{S_1(t)}{S_1(0)}) \end{pmatrix}$$

Giving this relation, we do as before thanks to the fact that S_i is always decreasing. Using a multidimensional substitution, we have:

$$T_1^2 = \int_0^{T_1}\int_0^{T_1} dtds \underset{(s_1,s_2)=(S_1(t),S_2(s))}{=} \int_{S_1(0)}^{S_1(T_1)}\int_{S_2(0)}^{S_2(T_1)} \frac{ds_1 ds_2}{(s_1,s_2)(-B(N-S)-\Gamma(\log(\frac{s_1}{S_1(0)}),\log(\frac{s_2}{S_1(0)}))^T}$$

And likewise:

$$T_2^2 = \int_{S_1(0)}^{S_1(T_2)}\int_{S_2(0)}^{S_2(T_2)} \frac{ds_1 ds_2}{(s_1,s_2)(-B(N-S)-\Gamma(\log(\frac{s_1}{S_1(0)}),\log(\frac{s_2}{S_1(0)}))^T}$$

$$T_1T_2 = \int_{S_1(0)}^{S_1(T_1)}\int_{S_2(0)}^{S_2(T_2)} \frac{ds_1 ds_2}{(s_1,s_2)(-B(N-S)-\Gamma(\log(\frac{s_1}{S_1(0)}),\log(\frac{s_2}{S_1(0)}))^T}$$

Therefore, we could have the access to the value of T_1 or T_2 provided we have the value of $(S_i(T_j))_{(i,j)\in\{1,2\}}$. Yet, this is non trivial question as the former method doesn't work here. We could find these value if we had access to the value of I at the peak, which is also unknown. Nonetheless, this method represents a great asset as it can be easily generalized to the multi-compartment model.

3.3 Approximating the time of peak in the SEIR case

In this section, we address Bacaër's conjecture [2] about the time of peak in the SEIR model.

3.3.1 Bacaër's conjecture

In the case of a SEIR model, Bacaër defines the time of peak as the time T where $E+I$ is maximal. In [2], it is proven that $S(T) = N\frac{K_{I,R}}{K_{I,S}}$ and that $E(T)+I(T) = N(1-\frac{K_{I,R}}{K_{I,S}}+\frac{K_{I,R}}{K_{I,S}}\log(N\frac{K_{I,R}}{K_{I,S}(N-1)}))$.

Let $M = \begin{pmatrix} -K_{E,I} & K_{I,S} \\ K_{E,I} & -K_{I,R} \end{pmatrix}$ and let λ_+ be the largest eigenvalue of M. Bacaër's conjecture reads

$$T(N) = \frac{\log N}{\lambda_+} + o(\log(N))$$

as $N \to +\infty$.

In [2], the following bound is proved

$$T(N) \geq \frac{\log N}{\lambda_+} + K$$

where K does not depend on N. The proof relies on a monotony argument.

3.3.2 Numerical testing of the conjecture

Fist, we tried to obtain the same type of graph as Figure 7 to observe the convergence of the estimate of the conjecture.

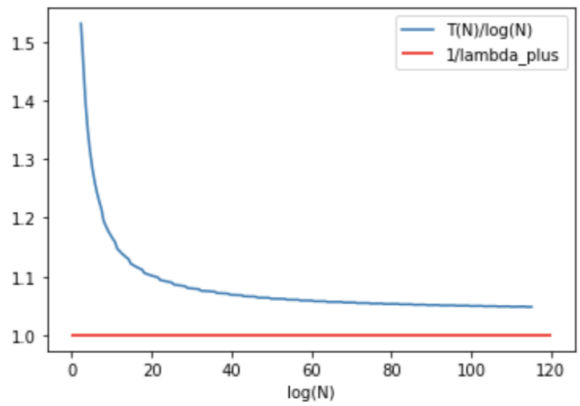

Figure 11: Plot of $T(N)/\log(N)$ w.r.t $\log(N)$

In Figure 11, Bacaër's lower bound can be verified. However, it is not clear whether the conjecture truly holds.

3.3.3 A different idea for the SEIR case

Let $\theta \in]0,1[$. We aim to control the behaviour of the SEIR variables by introducing $\tilde{S}, \tilde{E}, \tilde{I}, \tilde{R}$, the solutions to the system linearized around $S/N = \theta$.

Let $M_\theta = \begin{pmatrix} -K_{E,I} & \theta K_{I,S} & 0 \\ K_{E,I} & -K_{I,R} & 0 \\ 0 & K_{I,R} & 0 \end{pmatrix}$

The linearized system is

$$\frac{\mathrm{d}\tilde{X}}{\mathrm{d}t} = M_\theta \tilde{X}$$

where $\tilde{X} = (\tilde{E}, \tilde{I}, \tilde{R})^\top$.

First, we compute the eigenvectors and eigenvalues of M_θ. It suffices to use the 2×2 square matrix consisting of the first two rows and columns, which characteristic polynomial is

$$\chi(X) = X^2 + (K_{E,I} + K_{I,R})X + K_{E,I}(K_{I,R} - \theta K_{I,S})$$

The discriminant is

$$\Delta = (K_{E,I} - K_{I,R})^2 + 4\theta K_{I,S} K_{E,I}$$

There are two eigenvalues :

$$\lambda_{\pm,\theta} = \frac{1}{2}(-K_{E,I} - K_{I,R} \pm \sqrt{\Delta})$$

The largest eigenvalue is positive if and only if $\theta > \frac{K_{I,R}}{K_{I,S}}$.

The corresponding eigenvector basis is formed by

$$u_{\pm,\theta} = \begin{pmatrix} 1 \\ \frac{K_{E,I}}{\lambda_{\pm,\theta} + K_{I,R}} \\ \frac{K_{E,I} K_{I,R}}{\lambda_{\pm,\theta}(\lambda_{\pm,\theta} + K_{I,R})} \end{pmatrix}$$

Let A be the set of vectors $(S, \tilde{S}, E, \tilde{E}, I, \tilde{I})$ such that, and B the set of vectors $(S, \tilde{S}, E, \tilde{E}, I, \tilde{I})$ such that $S/N \leq \theta$. The idea is to use the fact that the set $A \cup B$ is invariant under the flow of the global differential system, when S, E, I, R follows the true SEIR dynamics and $\tilde{S}, \tilde{E}, \tilde{I}, \tilde{R}$ the linearized one. This invariance can be proven for example by using a characterization by Brezis [3].

Let $T_\theta, \tilde{T}_\theta$ be respectively the time when $S \leq \theta N$ and $\tilde{S} \leq \theta N$ (recall that, with our initial conditions, S and $\tilde{S}$ start above θN and then decreases). By the flow invariance property we have $T_\theta \leq \tilde{T}_\theta$. Now, one has to compute the time $\tilde{T}_\theta$.

Let $\alpha, \beta > 0$ such that $\beta u_+ \leq \tilde{X}(0) \leq \alpha u_+$ (inequalities are element-wise). As the flow of the equation is nondecreasing, one has $\beta e^{\lambda_{+,\theta}} u_+ \leq \tilde{X}(t) \leq \alpha e^{\lambda_{+,\theta}} u_+$. Summing the three inequalities and using the fact that when $S/N = \theta$, $E + I + N = N(1 - \theta)$, it is possible to control the time $\tilde{T}_\theta$. Our final idea was to note that at the time of the peak, the value of $E + I + R$ is known [2], which enables to find a time that matches the peak with the moment when $\tilde{S}/N = \theta$. However, this corresponds to a zero eigenvalue, whereas we expected the eigenvalue $\lambda_{+,1}$ to match Bacaër's conjecture.

4 Conclusion

In this project, we studied a few properties of three epidemiological models : SIR, SEIR and bi-compartmental. In order to find tropical approximations of the trajectories in such models, we studied numerically some properties of the amoebas of these systems. We also had to find approximations for time of peaks in these models. We improved Bacaër's [1] bound on the time of peak in a SIR model, and we tried to find similar results in the SEIR case, using techniques involving flow invariance and Perron-Frobenius vectors.

References

[1] N. Bacaër. Sur le pic épidémique dans un modèle sir. 2020.

[2] N. Bacaër, F. Hamelin, and H. Inaba. De nouvelles propriétés du pic épidémique. *Quadrature*, 119, 2021.

[3] H. Brezis. On a characterization of flow-invariant sets. *Communications on Pure and Applied Mathematics*, 23(2):261–263, 1970.

[4] S. Gaubert, M. Akian, X. Allamigeon, M. Boyet, B. Colin, T. Grohens, L. Massoulié, D. P. Parsons, F. Adnet, É. Chanzy, et al. Understanding and monitoring the evolution of the covid-19 epidemic from medical emergency calls: the example of the paris area. *Comptes Rendus. Mathématique*, 358(7):843–875, 2020.

[5] W. O. Kermack and A. G. McKendrick. A contribution to the mathematical theory of epidemics. *Proceedings of the royal society of london. Series A, Containing papers of a mathematical and physical character*, 115(772):700–721, 1927.

[6] V. Noël, D. Grigoriev, S. A. Vakulenko, and O. Radulescu. Tropical Geometries and Dynamics of Biochemical Networks Application to Hybrid Cell Cycle Models. *Electronic Notes in Theoretical Computer Science*, 284:75–91, 2012. Proceedings of the 2nd International Workshop on Static Analysis and Systems Biology (SASB 2011).